

Advanced Machine Learning

Mid-Semester Examination

3rd Year, Semester 6

NST, Rishihood University, Sonipat

Date: 1 March 2026

PART A: RESEARCH PAPER SELECTION

Release Date: 1 March 2026, 7:00 PM

Deadline: 3 March 2026, 11:59 PM

Marks: 50

Mandatory Requirement:

Only students who successfully complete **Part A** will be allowed to attempt **Part B** and **Part C**.

Objective

The objective of Part A is to ensure that each student takes deep ownership of **one classical machine learning method** before proceeding to reproduction, experimentation, and explanation in Parts B and C.

Each student must select **exactly ONE research paper** and work only on this paper for the remainder of the examination in Part B and Part C.

What You Must Select

You must select exactly **ONE research paper** from the **classical machine learning era (2009–2012)** whose primary technical contribution is based on at least one of the following:

- ARIMA or time-series modeling and variants
- Gaussian Mixture Models (GMM), EM-based clustering, or variants
- Support Vector Machines (SVM), kernel methods, or variants

The same paper will be used for Parts B and C.

Paper Selection Constraints (Mandatory)

Publication Venue (Strict)

The paper must be published or accepted in a **CORE A* conference or journal (2009–2012 editions)**.

Reference: <https://portal.core.edu.au/conf-ranks/?search=&by=all&source=CORE2010&sort=arank&page=1>

Allowed venues include:

- **Core Machine Learning:** NeurIPS, ICML, JMLR, AISTATS, PAMI

- **General AI:** AAAI, IJCAI
- **Computer Vision:** CVPR, ICCV, ECCV
- **Speech and NLP:** ACL, EMNLP
- **Data Mining and Databases:** KDD, ICDM, ICDE

Only papers whose **primary contribution aligns with classical ML methods** (ARIMA, GMM, SVM, or close variants) are allowed.

Workshop papers, theses, blogs, and unpublished manuscripts are **NOT allowed**.

Year

The paper must be published between **2009 and 2012** (inclusive).

Nature of Contribution

The primary contribution must be **methodological** or an **application** of the above algorithms.

Papers whose main contribution is only a dataset, benchmark, or system are **NOT allowed**.

Acceptance Status

arXiv-only or unaccepted papers are **NOT allowed**.

Reproducibility and Compute Constraints (Very Important)

Students are fully responsible for checking whether the selected paper can be reasonably reproduced.

Strong recommendations and requirements:

- Dataset and/or code availability is recommended (not mandatory)
- At least one later exam question will involve inference
- Experiments must be runnable on a standard personal laptop (CPU-only)
- GPU-dependent or heavy-compute setups are explicitly forbidden
- Implementations compatible with **scikit-learn** or equivalent classical ML libraries are preferred

The dataset used (or a suitable substitute) must satisfy at least one of the following conditions, as applicable to the selected algorithm:

- At least 50 consecutive time points
- At least 2 continuous numeric features
- At least 1 usable label (or clearly derivable label)

You may use re-implementation, toy datasets, or simplified experimental setups.

Code obsolescence is NOT an excuse.

Do NOT Select Papers That

- Require special approvals or restricted access
- Use proprietary, medical, clinical, telecom, or industry data
- Depend on unavailable or licensed datasets
- Require GPUs or large-scale infrastructure

Submission Mechanism and Uniqueness (Strict)

Paper selection must be submitted via the official Google Form:

<https://forms.gle/mvcCy88gRDfWrxck6>

A live, read-only response sheet is available here:

https://docs.google.com/spreadsheets/d/17_pjq5yqDEG0t4J6qS7KE_wm8-xsmF42xYKm2cPSCBM/edit

This is a **first-come, first-select** process.

Rules

- One paper per student
- Duplicate selections are **NOT allowed**
- Once submitted, the paper **CANNOT be changed**
- Students must check the response sheet before submission

If two students submit the same paper, only the student with the **earliest timestamp** will be allowed to proceed to Parts B and C.

All later duplicate submissions will be rejected without exception.

Example: How to Select a Good Paper

Example (Application-Based, SVM):

- **Paper:** “Object Detection with Discriminatively Trained Part-Based Models” (2009)
- **Authors:** Pedro Felzenszwalb, Ross Girshick, David McAllester, Deva Ramanan
- **Venue:** IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI)

Why it works:

- Clear real-world application: object detection in images
- Core method is classical machine learning (latent SVM)
- Uses well-understood feature representations (HOG) and linear SVMs
- Widely available datasets (e.g., PASCAL VOC) with smaller subsets usable for reproduction

Why this paper is appropriate for Part A:

- The contribution is methodological *and* application-driven
- Experiments can be simplified and run on CPU-only setups
- The learning objective and inference procedure are explicitly described
- Failure cases (viewpoint changes, occlusion, part deformation) are clearly observable

What a student should check before selecting such a paper:

- Can I run or reimplement a simplified version of the model using a small dataset?
- Can I explain how SVMs are used for detection rather than classification?
- Can I reason about when and why the detector fails?

What to Avoid (Strongly Recommended)

Avoid papers that have:

- Licensed or restricted datasets
- Missing hyperparameter or experimental details
- Custom kernels or solvers unavailable in standard libraries
- Extremely large datasets (millions of rows)
- Results dependent on undocumented train/test splits
- Heavy theory with no empirical validation

Important Notes (Read Carefully)

- Paper selection is final once submitted
- No paper change requests will be entertained
- Using LLMs to understand papers is allowed
- Copying text without understanding will negatively impact Parts B and C
- Part C (proctored) will test understanding of the exact paper without internet access
- If you cannot explain your own paper without tools, you will lose marks
- In case of duplicate selection, only the earliest valid submission is accepted
- All other duplicate entries will be automatically disqualified from Parts B and C

The Google Form will remain open for **48 hours**.

Failure to comply with any Part A rule may result in **disqualification from Parts B and C** and loss of marks for Part A.

LLM Usage Disclosure and JSON Submission (Mandatory)

The use of Large Language Models (LLMs) such as ChatGPT, Claude, Gemini, or equivalent tools is **explicitly allowed** for Part A of this examination.

However, **full, honest, and structured disclosure of all LLM usage is mandatory**.

LLMs may be used as:

- Reading and comprehension aids
- Concept clarification tools
- Assistance in identifying assumptions, limitations, and failure cases

LLMs are **not** a substitute for understanding. Students are fully responsible for all submitted content and must be able to explain it independently in Part C.

Submission Mechanism

Along with the Part A paper selection Google Form, each student must create a **public GitHub repository** with the prescribed naming convention and include their complete LLM usage disclosure within the repository.

Submission rules:

- Repository name must be in the format: `<rollnumber-midsem>`
- The repository must be **public** at the time of submission
- A single JSON file named `llm_usage_partA.json` must be present at the repository root
- Partial, selective, or edited disclosure of LLM usage is **not allowed**

Students must submit the **GitHub repository URL** via the official Google Form.

Failure to create the repository correctly, missing the required file, or providing an invalid or incomplete LLM usage disclosure may result in penalties or disqualification from Parts B and C.

Required JSON Structure

The JSON file must follow the structure below.

1. Student Metadata

```
"student_metadata": {  
  "name": "Student Name",  
  "roll_number": "23NST045",  
  "course": "Advanced Machine Learning",  
  "exam": "Mid-Semester Examination",  
  "part": "Part A",  
  "submission_date": "YYYY-MM-DD"  
}
```

2. LLM Tools Used

All LLM tools used during Part A must be declared.

```
"llm_tools_used": [
  {
    "tool_name": "ChatGPT",
    "model": "GPT-4",
    "provider": "OpenAI"
  }
]
```

3. Complete LLM Interaction Log (Mandatory)

Students must disclose **all LLM interactions** used for paper selection, understanding, analysis, or planning.

```
"full_llm_interaction_log": [
  {
    "interaction_id": 1,
    "date": "YYYY-MM-DD",
    "tool_name": "ChatGPT",
    "model": "GPT-4",
    "purpose": "Understanding the EM objective in GMM",
    "prompt": "Explain the EM log-likelihood for Gaussian Mixture Models.",
    "llm_response_used": "Partially",
    "how_it_helped": "Clarified intuition before re-reading the paper.",
    "student_verification": "Verified using equations from the paper and class notes.",
    "confidence_level": 4
  }
]
```

Notes:

- `prompt` must be the exact text entered (verbatim)
- `llm_response_used` must be one of: Yes / No / Partially
- `student_verification` is compulsory and must be written in the student's own words
- `confidence_level` must be an integer from 1 (low) to 5 (high)

4. Top 5 Most Valuable Prompts

From the complete interaction log, students must select the **five prompts** they believe contributed the most to their understanding. Select prompts that genuinely influenced your understanding or decisions during the selection of the paper; reflective and feasibility-oriented prompts are preferred.

```
"top_5_prompts": [
  {
    "rank": 1,
    "interaction_id": 2,
    "prompt": "What are common failure cases of EM-based GMM clustering?",
    "why_important": "Helped identify sensitivity to initialization as a key limitation."
  }
]
```

```
}  
]
```

Each selected prompt must:

- Appear in the full interaction log
- Be ranked from 1 (most impactful) to 5
- Include a justification written by the student

5. Student Declaration (Mandatory)

```
"student_declaration": {  
  "statement": "I declare that this JSON file contains a complete and honest record  
of my LLM usage for Part A.",  
  "understanding_acknowledged": true,  
  "signature": "Student Name",  
  "date": "YYYY-MM-DD"  
}
```

Important Academic Integrity Notes

- Honest disclosure of LLM usage will **not reduce marks**
- Failure to disclose or selective disclosure will be treated as academic misconduct
- Students must be able to explain all submitted work without LLM assistance in Part C
- Any mismatch between disclosed usage and demonstrated understanding may result in penalties

Evaluation Scheme for Part A

Part A will be evaluated out of **50 raw marks** (scaled to 5% of the total mid sem marks).

Part A evaluates the correctness and feasibility of the selected paper, along with the completeness and quality of the student's LLM usage disclosure. Conceptual understanding, assumptions, and limitations will be evaluated in Parts B and C.

Evaluation Criterion	Marks	Detailed Rubric
Paper Validity: Venue	5	Correct CORE A* venue (2009–2012)
Paper Validity: Year	4	Published between 2009–2012 (inclusive)
Paper Validity: Method Alignment	4	Primary contribution aligns with classical ML methods (ARIMA / GMM / SVM or close variants)
Paper Validity: Paper Type	2	Not a dataset, benchmark, system paper, or restricted-access work
Reproducibility: Dataset Availability	5	Dataset is publicly available or has a clearly usable substitute
Reproducibility: Compute Feasibility	5	Experiments are feasible on a standard CPU-only personal laptop
Reproducibility: Experimental Scope	5	Scope is reasonable for student-level reproduction or simplification
LLM Disclosure: JSON Validity	4	LLM usage JSON file is valid, readable, and follows the prescribed format
LLM Disclosure: Completeness	3	All LLM tools and interactions related to Part A are disclosed
LLM Disclosure: Verification	3	Verification statements and student declaration are present
Top-5 Prompts: Analytical Depth	6	Prompts demonstrate analytical intent (why, when, failure modes, trade-offs)
Top-5 Prompts: Relevance	4	Prompts are clearly relevant to paper selection and feasibility assessment
Total	50	

Important Notes:

- Part A does not evaluate conceptual mastery of the paper.
- Honest and complete disclosure of LLM usage will not reduce marks.
- Incomplete or misleading disclosure may result in penalties or disqualification from Parts B and C.
- Deeper understanding of the selected paper will be assessed in Parts B and C.

End Note

No submissions will be accepted after the stated deadline under any circumstances.

All the best.

Penalty for Incorrect Submission [Update 1 March 2026, 9:52 PM]

Based on requests received from students, multiple submissions have been enabled to allow corrections in case of genuine mistakes.

You may resubmit if needed; however, only the **latest submission** will be evaluated for Part A/ Part B and Part C.

A **20% deduction in the Part A score** will be applied for **each additional submission beyond the first**.

Students are strongly advised to avoid unnecessary resubmissions and ensure that all guidelines are carefully followed before submitting.